Simple CMOS Process

	process	Real step	Simulation notes	extract
00	Wafer prepare		In simulation, thickness of silicon is 20um	
1.	Wafer check		P type <100> 60Ω.cm	
2.	Nwell oxidation 110±10nm	3# Clean Oxidation into furnace Oxidation outof furnace	In simulation， deposit 110nm SiO2	
3.	Nwell Photolithography mask: NWELL	Pretreatment Photo deposit Exposure Development Check	In simulation： 1. Deposit photo 1.2um 2. Etch photo with the correct shape (shape for Nwell should be obtained in the 2D profile of the device)	
4.	Etch SiO2	Etch SiO2		
5.	N well implant	N well implant	P31(phosphorus) 120Kev 5E12 If you use deckbuild menu to generate the codes, pls. Delete the definition for tilt and rotation	Junction depth of Nwell
6.	Remove all of the photoresist	Etch photo		
7.	Drive-in Nwell 3-2# NWELL	3# Clean Oxidation into furnace Oxidation outof furnace	In order to get the step height of silicon surface, you need to get the y.val for silicon boundaries inside and outside the Nwell. 0-9.5um is outside of the Nwell, 9.5-16.5um is inside the Nwell. When you do extract, x.val should be far away from 9.5. min.bound is used to get the y.val of the material surface. max.bound is used to get the y.val for the material bottom This parameter can not be generated by the menu, pls. Type down by yourself.	1. Thickness of SiO2 inside Nwell 2. Thickness of SiO2 outside Nwell 3. Step height of silicon surface 证明：内外两处… 4. Junction depth of Nwell
8.	Remove all of SiO2	Wet etch SiO2		

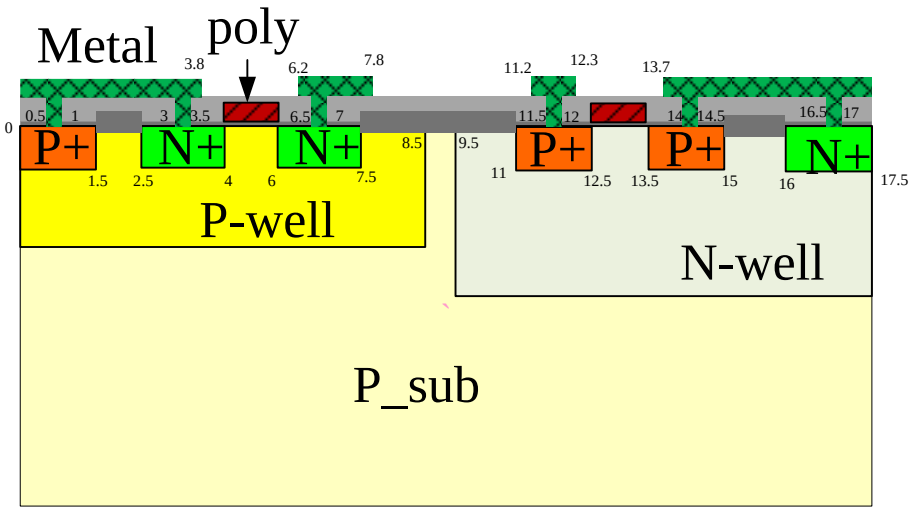
Simple CMOS Process

NWELL 3-2 Drive-in Nwell								
Step	Temp °C	Ramp °C/min	Time	N2(slpm)	O2(slpm)	H2(slpm)	Tca (sccm)	Boat Cm/min
Push in	800		15’		8			
Ramp to	1180		60’		8			
Anneal	1180		200’	8	4			
Ramp to	800		100’	8				
Push out	800		15’	6				

Note：1. Temperature can't change suddenly. t.final is needed when temperature change；

2. For O2(slpm)8, code is f.o2=8（In deckbuild menu for diffuse, choose GasFlow to set f.o2）

3. When you use deckbuild menu to generate codes with t.final, there may be a grammar error. If it has an error, check the codes and add a space when needed.



NWELL PWELL ACTIVE NPLUS PPLUS POLY CONT METAL